The small-parameter power for higher-dimensional Liouville first passage percolation

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Abstract

For each fixed $d \geq 2$, let $\lambda_d(\xi) = 1 - \xi Q_d(\xi)$ be the distance exponent for Liouville first passage percolation associated with the whole-space log-correlated Gaussian field on $\mathbb{R}^d$, normalized to have covariance singularity $\log |x-y|^{-1}$. We prove that $\lambda_d(\xi) \leq C_d \xi^{4/3}$ for sufficiently small $\xi > 0$. Together with the planar lower bound and restriction to a two-dimensional subspace, this gives $\lambda_d(\xi) = \xi^{4/3+o(1)}$. The upper bound follows from consecutive partitions of arbitrary paths and a Gaussian estimate for each possible sequence of recorded subpath data. Correlations between scales are controlled directly; independence along the path is not assumed. We also compute the transverse Gaussian limit and compare its variational constants in different dimensions. We do not identify a leading LFPP coefficient or remove the logarithm from the lower bound.

Research manuscript. Internally checked; not independently refereed.

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1 Introduction and main theorem

Fix an integer $d \geq 2$. All norms on $\mathbb{R}^d$ are Euclidean. Fix a nonnegative, radially symmetric function $\kappa \in C_c^\infty(\mathbb{R}^d)$ with $\|\kappa\|_{L^2} = 1$. Let W be white noise on $\mathbb{R}^d \times (0,\infty)$, with Lebesgue control measure, and set

$$K_t(z) = t^{-(d+1)/2}\kappa(z/t), \qquad H_N(x) = \int_{2^{-N}}^1 \int_{\mathbb{R}^d} K_t(x-y)\, W(\,dy,\, dt). \tag{1.1}$$

We use a smooth modification of H_N. In particular, $\operatorname{Var} H_N(x) = N \log 2$. Put $U = (-2,2)^d$ and

$$D_N^\xi = \inf_{P:0\to e_1,\ P\subset U} \int_P e^{\xi H_N(z)}|dz|, \qquad a_N(\xi) = \operatorname{median} D_N^\xi. \tag{1.2}$$

The paths are continuous and rectifiable, and arclength is counted with multiplicity. The infimum is unchanged if one restricts to piecewise C^1 paths, by polygonal approximation on the compact image of each path. One may further restrict to polygons with rational internal vertices, which makes the infimum a measurable function of the continuous field. The same convention applies to internal distances below.

By [4, Proposition 1.1], the deterministic exponent

$$\lambda_d(\xi) = -\lim_{N\to\infty} \frac{\log a_N(\xi)}{N \log 2} = 1 - \xi Q_d(\xi) \tag{1.3}$$

exists. When writing constants with subscript d, we choose one such kernel in each dimension. The use of a fixed compactly supported kernel places (1.1) exactly in the approximation scheme of that paper. The comparison in Appendix A also shows that the exponent is independent of this kernel. For small ξ, this is the exponent in the weak exponential metric setup of [2]; see its Proposition 1.10 and the discussion following it. The field is the ordinary GFF only when $d = 2$.

Theorem 1.1 (Main theorem)**.** *For each fixed $d \geq 2$, there exist constants $c_d, C_d > 0$ and $\xi_d \in (0, e^{-1})$ such that*

$$c_d \frac{\xi^{4/3}}{\log(1/\xi)} \leq \lambda_d(\xi) \leq C_d \xi^{4/3}, \qquad 0 < \xi < \xi_d. \tag{1.4}$$

Consequently,

$$\boxed{\lim_{\xi\downarrow 0} \frac{\log \lambda_d(\xi)}{\log \xi} = \frac{4}{3}.} \tag{1.5}$$

The estimate proved here is the upper bound in (1.4). The lower bound follows from the planar bound of Ding–Goswami [3] and restriction to a plane. This restriction comparison is also the content of [1, Lemma 1.4]; we include a comparison for the precise point-to-point cutoff used here in Appendix A. No claimed leading-coefficient result from an earlier working draft is used.

The statement is for fixed d. There is no assertion that C_d or ξ_d^{-1} is bounded as $d \to \infty$. Also, (1.5) is a logarithmic asymptotic, not the assertion that $\lambda_d(\xi)/\xi^{4/3}$ converges. The latter assertion is formulated separately in Section 6.

1.1 The corresponding metric dimension

For the weak γ-exponential metrics of [2], write d_γ for their almost sure Hausdorff dimension. The parameter correspondence is

$$Q_d(\xi) = \frac{d}{\gamma} + \frac{\gamma}{2}, \qquad \xi = \frac{\gamma}{\mathsf{d}_\gamma}, \qquad 0 < \gamma < \sqrt{2d}; \tag{1.6}$$

see [2, (1.5) and Corollary 1.9].

Corollary 1.2. *For each fixed $d \geq 2$, there are positive constants c'_d, C'_d, γ_d such that*

$$c'_d \frac{\gamma^{4/3}}{\log(1/\gamma)} \leq \mathsf{d}_\gamma - d \leq C'_d \gamma^{4/3}, \qquad 0 < \gamma < \gamma_d. \tag{1.7}$$

In particular,

$$\lim_{\gamma \downarrow 0} \frac{\log(\mathsf{d}_\gamma - d)}{\log \gamma} = \frac{4}{3}. \tag{1.8}$$

1.2 Relation to previous work and scope

Ding–Gwynne–Zhuang ask for the small-parameter asymptotics of $1 - \xi Q_d(\xi)$ in [4, Problem 7.4]. The upper bound in [1, Theorem 1.3] is $\lambda_d(\xi) \leq \xi\sqrt{2d-2}$. Their discussion after Theorem 1.2 records the planar lower bound of order $\xi^{4/3}/\log(1/\xi)$. Thus the upper estimate in Theorem 1.1, if confirmed, is a new improvement over the bounds in these sources. A literature search is not a verification of priority; Appendix B records the scope of the checks made for this manuscript.

The proof does not require the construction or uniqueness of a limiting metric. It uses the established distance exponent only after a bound simultaneous over all paths has been obtained. The independent-scale representation in (1.1) is used for covariance calculations, not to declare the fields seen at successive path segments independent.

Throughout the proof, C denotes a finite deterministic constant which may change between occurrences. It may depend on d and the fixed kernel κ, but not on N, ξ, or the geometric scale. A dependence on another fixed parameter is indicated. We fix the subdivision parameter

$$M = 16 \tag{1.9}$$

once and for all. Thus constants depending on M are included in C.

2 Gaussian estimates for curve measures

For a rectifiable curve Q of positive length, let ν_Q denote normalized arclength measure, with multiplicity. We say that a probability measure μ satisfies the growth bound at scale $R > 0$ with constant B if

$$\mu(B(z,s)) \leq B \min(1, s/R), \qquad z \in \mathbb{R}^d, \quad s > 0. \tag{2.1}$$

Here and below $B(z,s)$ denotes a Euclidean ball.

Lemma 2.1 (Transport estimate)**.** *Let $R \geq r > 0$. Suppose μ_R, ν_R satisfy (2.1) at scale R and μ_r, ν_r satisfy it at scale r, all with constant B. Suppose the first pair admits a coupling of displacement at most u and the second a coupling of displacement at most v. Set $\sigma_R = \mu_R - \nu_R$ and $\sigma_r = \mu_r - \nu_r$. Then, uniformly in N,*

$$\left|\mathrm{Cov}\left(\langle H_N, \sigma_R\rangle, \langle H_N, \sigma_r\rangle\right)\right| \leq C_B \frac{\sqrt{uv}}{R}. \tag{2.2}$$

In particular, $\mathrm{Var}\langle H_N, \mu_R - \nu_R\rangle \leq C_B u/R$.

Proof. Smoothness and compact support of κ give

$$\|\sigma_R * K_t\|_\infty \leq \frac{C_B}{R} t^{-(d-1)/2} \min(1, u/t), \tag{2.3}$$

$$\|\sigma_r * K_t\|_1 \leq C t^{(d-1)/2} \min(1, v/t). \tag{2.4}$$

For the first estimate without the displacement factor, bound K_t by $Ct^{-(d+1)/2}$ on a ball of radius Ct, and use (2.1) for each measure. If $u \le t$, integrate the gradient of K_t along the displacement coupling. The interpolated measures have the same growth bound at radii comparable to t, since their points lie within distance $u \le t$ of points of μ_R. This gives the additional factor u/t. The second estimate follows from $\|K_t\|_1 = Ct^{(d-1)/2}$ and $\|\nabla K_t\|_1 = Ct^{(d-3)/2}$.

The white-noise isometry and (2.3)–(2.4) imply that the covariance is at most

$$\frac{C_B}{R}\int_0^\infty \min(1, u/t)\min(1, v/t)\,dt.$$

If $a = \min(u, v) > 0$ and $b = \max(u, v)$, the integral is $a(2+\log(b/a)) \le 2\sqrt{ab}$. If either displacement is zero, the corresponding signed measure vanishes. This proves the lemma. Notice that the powers of t from the ambient dimension cancel. □

Lemma 2.2 (Stability of normalized arclength)**.** *Let Q, Q' be polygons with the same number $m \le M$ of edges and corresponding vertices z_i, z_i'. Suppose their edge lengths lie in $[cR, CR]$ and their total lengths are at most CR. If $\eta = \max_i |z_i - z_i'|$, their normalized arclength measures admit a coupling of displacement at most $C_{c,C,M}\eta$. The analogous bound for two chords has constant one.*

Proof. Parametrize both polygons at constant speed on $[0, 1]$. Their normalized cumulative edge lengths differ by at most $C_{c,C,M}\eta/R$. Let ϕ be the piecewise affine increasing map which sends the vertex times of Q to those of Q'. Then $\|\phi - \mathrm{id}\|_\infty \le C_{c,C,M}\eta/R$. Corresponding affine interpolation gives $|Q(t) - Q'(\phi(t))| \le \eta$, while the speed of Q' is at most CR. Consequently $\sup_t |Q(t) - Q'(t)| \le C_{c,C,M}\eta$. Use the coupling given by a uniform $t \in [0, 1]$. □

We use the following two standard Gaussian estimates. If $(X_z)_{z\in T}$ is a centered separable Gaussian process with canonical distance d_X, then

$$\mathbb{E}\sup_{z\in T}(X_z - X_{z_0}) \le C\int_0^{\operatorname{diam}(T,d_X)} \sqrt{\log\left(2\mathcal{N}(T, d_X, s)\right)}\,ds, \tag{2.5}$$

where $z_0 \in T$ and $\mathcal{N}$ is a covering number. If $T_* = \sup_{z\in T} X_z$ and $\sup_z \operatorname{Var} X_z \le \sigma^2$, then

$$\mathbb{E}e^{tT_*} \le \exp\left(t\mathbb{E}T_* + \frac{t^2\sigma^2}{2}\right), \qquad t \ge 0. \tag{2.6}$$

The first estimate is the Gaussian entropy bound of [5]. For the second, a finite supremum of Gaussian linear functionals is σ-Lipschitz on the underlying Gaussian space. The Gaussian logarithmic Sobolev inequality and the Herbst argument give (2.6); see [6]. Separable approximation proves the version used here. All applications below have a finite-dimensional compact parameter space and a finite entropy integral.

Lemma 2.3 (The root and terminal errors)**.** *Let $I_0 = [0, e_1]$, let $\varepsilon = 2^{-N}$, and set*

$$\omega_N = \sup\{|H_N(z) - H_N(w)| : z, w \in \overline{U},\ |z - w| \le 8\varepsilon\}. \tag{2.7}$$

Then $\langle H_N, \nu_{I_0}\rangle = O_{\mathbb{P}}(1)$ and $\omega_N = O_{\mathbb{P}}(\sqrt{N})$.

Proof. The covariance kernel at scale t is $t^{-1}(\kappa * \kappa)((z - w)/t)$, which is bounded in absolute value by $Ct^{-1}\mathbf{1}_{\{|z-w|\le Ct\}}$. The variance of the average over the unit segment is therefore at most $C\int_0^1 t\ dt/t = C$.

Also $\sup_x \mathbb{E}|\nabla H_N(x)|^2 \le C\varepsilon^{-2}$. Each difference in (2.7) has bounded variance. For the difference process indexed by (z, w), a Euclidean parameter mesh of size $c\varepsilon s$ gives a canonical mesh of size s. For $s \in (0, 1)$ its covering number is at most $C\varepsilon^{-2d}s^{-2d}$. Formula (2.5), applied also to the negative process, bounds $\mathbb{E}\omega_N$ by $C\sqrt{N}$. Markov's inequality gives the claimed bound in probability. □

3 A consecutive partition of every path

Fix a continuous rectifiable path P from 0 to e_1 in U. A node u of the partition has a consecutive subpath P_u, initial and final points x_u, y_u, and chord length $R_u = |y_u - x_u| > 0$. The root is the full path and has $R_0 = 1$.

At an internal node, let $r = R_u/M$. Starting from its initial time, successively cut the subpath at the first time its displacement from the current starting point is r. Every completed piece has chord length exactly r. There is at least one completed piece because $R_u > r$. If a final residual subpath remains, retain it when its chord length is greater than $r/2$, and otherwise merge it with the last completed piece. A constant terminal piece may simply be discarded. This convention also handles a nonconstant residual subpath whose endpoints agree.

Every resulting child v has

$$\frac{R_u}{2M} \le R_v \le \frac{3R_u}{2M}, \tag{3.1}$$

and every edge except possibly the last has length R_u/M. Each child subpath is contained in $\overline{B}(x_v, 4R_v)$: a completed or retained residual piece lies in a ball of radius r about its start, while a merged piece lies in one of radius $2r$. The lower bound $R_v \ge r/2$ proves the assertion. The root satisfies

$$P_u \subset \overline{B}(x_u, C_d R_u), \qquad C_d = \max(4, 2\sqrt{d}), \tag{3.2}$$

and every nonroot node satisfies this with constant 4.

Stop when $R_u \le \varepsilon = 2^{-N}$. The tree is finite. Indeed, each node has finitely many first-exit pieces since the original path has finite length, and (3.1) bounds the depth. Every root-to-leaf chain has at most

$$J_N = 1 + \left\lceil \frac{N}{\log_2(2M/3)} \right\rceil \tag{3.3}$$

internal nodes. No bound on the total number of nodes is needed.

For an internal node, define

$$S_u = \sum_{v \text{ child of } u} R_v, \quad a_u = \log(S_u/R_u), \quad p_{uv} = R_v/S_u, \quad H_u = \langle H_N, \nu_{[x_u,y_u]} \rangle. \tag{3.4}$$

The triangle inequality gives $a_u \ge 0$. Write

$$G_u = H_u - \sum_v p_{uv} H_v. \tag{3.5}$$

Let Q_u be the polygon of the ordered child chords. Then $G_u = \langle H_N, \nu_{[x_u,y_u]} - \nu_{Q_u} \rangle$. Define a unit flow by $\theta(0) = 1$ and $\theta(v) = \theta(u)p_{uv}$.

Lemma 3.1 (Deterministic lower bound)**.** *For every path and its partition,*

$$\log \int_P e^{\xi H_N(z)} |dz| \ge \xi H_0 + \sum_{u \text{ internal}} \theta(u)(a_u - \xi G_u) - \xi \omega_N. \tag{3.6}$$

Proof. A leaf subpath has chord length $R_v \le \varepsilon$ and is contained in the ball of radius $4R_v$ about its start. Its points are within 5ε of every point of its chord. Its cost is therefore at least $R_v e^{\xi H_v - \xi\omega_N}$. The identities

$$R_v = \theta(v) \exp\left(\sum_{u \prec v} a_u \right), \qquad \sum_{v \text{ leaf}} \theta(v) H_v = H_0 - \sum_{u \text{ internal}} \theta(u) G_u$$

follow by telescoping along the tree; $u \prec v$ denotes an internal strict ancestor. Sum the leaf lower bounds, apply Jensen's inequality with the probability weights $\theta(v)$, and use these identities. □

3.1 Nodes with small excess

Fix a sufficiently small positive constant a_0, depending only on M. Call a node *good* when $a_u \le a_0$, and *bad* otherwise.

Lemma 3.2 (Geometry at a good node)**.** *The constant a_0 can be chosen so that every good node has exactly M children. Their chord polygon is a graph in the parent direction, and every edge makes an angle at most $1/10$ with that direction. Its normalized arclength measure and the parent chord measure satisfy* (2.1) *at scale R_u with a fixed constant, and admit a coupling of displacement at most $CR_u\sqrt{a_u}$. If z_i is the ith polygon vertex, then*

$$\left| z_i - \left(x_u + \frac{i}{M}(y_u - x_u) \right) \right| \le CR_u\sqrt{a_u}, \qquad 0 \le i \le M. \tag{3.7}$$

Proof. Write ℓ_i for the child chord lengths, t_i for their unit directions, and $e = (y_u - x_u)/R_u$. Then

$$\sum_i \ell_i |t_i - e|^2 = 2(S_u - R_u). \tag{3.8}$$

Since $\ell_i \ge R_u/(2M)$, each $|t_i - e|^2$ is at most $4M(e^{a_u} - 1)$. Taking a_0 sufficiently small gives the angle bound.

We may also require $e^{a_0} < 1 + 1/(4M)$. Put $r = R_u/M$ and let m be the number of children. If the residual was merged, then $S_u \in [(m - 1/2)r, (m + 1/2)r]$. Since $S_u \in [Mr, Mr + r/4)$, this forces $m = M$. A retained residual would instead give $S_u \in ((m - 1/2)r, mr)$, which is incompatible with the same interval: $m \le M$ gives $S_u < Mr$ and $m \ge M + 1$ gives $S_u > (M + 1/2)r$. With no residual, $S_u = mr$ and again $m = M$.

For $i < M$, the first i lengths equal R_u/M. Apply Cauchy–Schwarz to $\sum_{j \le i} \ell_j(t_j - e)$ and use (3.8) to obtain (3.7). The endpoint cases are immediate. Positivity of all longitudinal velocities gives the growth bound: the preimage of a ball of radius s has normalized arclength at most Cs/R_u.

For the coupling, parametrize Q_u at constant speed on $[0, 1]$. Its derivative has magnitude S_u and mean $R_u e$. Thus

$$\int_0^1 |Q_u'(t) - R_u e|^2 \, dt = S_u^2 - R_u^2.$$

The normalized parametrizations of the polygon and its chord consequently differ everywhere by at most $\sqrt{S_u^2 - R_u^2} \le CR_u\sqrt{a_u}$. This gives the stated coupling. □

4 Recording a chain and estimating one node

For the rest of the upper-bound proof set

$$\delta = \xi^{2/3}, \qquad \xi = \delta^{3/2}. \tag{4.1}$$

We assume $0 < \delta < \delta_0$ for a sufficiently small fixed δ_0. For each chord length R, let L be the unique dyadic number with $L/2 < R \le L$. At scale L use the fixed grid of half-open cubes of side δL in $\mathbb{R}^d$. The assigned cube of a point is unique.

If a parent and child have dyadic scales L, L', respectively, (3.1) implies

$$L'/L \in \{2^{-3}, 2^{-4}, 2^{-5}\}. \tag{4.2}$$

In particular, the dyadic scale indices along every chain strictly increase.

For a good node, record its parent scale and endpoint cubes, the bin

$$k = \lfloor a_u/\delta^2 \rfloor, \tag{4.3}$$

the index of the child followed by the chain, the next dyadic scale, and the two endpoint cubes of that child in the next grid. The other child vertices are not recorded.

For a bad node, choose a child v_* maximizing $H_u - H_v$, with any deterministic tie-breaking convention for its occurrence. Put

$$G_u^* = H_u - H_{v_*}. \tag{4.4}$$

Record the endpoints of this adverse child in the parent grid, and the endpoints of the followed child in the next grid, together with its scale. No occurrence index is recorded at a bad node. The number of children may be arbitrarily large, but all recorded endpoints lie in the fixed parent ball (3.2).

For a good node put $\widehat{G}_u = G_u$, and for a bad node put $\widehat{G}_u = G_u^*$. Define

$$Y_u = \delta^{-1/2}\widehat{G}_u, \qquad b_u = \begin{cases} k, & u \text{ good, in bin } k, \\ a_0/\delta^2, & u \text{ bad.} \end{cases} \tag{4.5}$$

Since $G_u \le \max_v(H_u - H_v)$, both kinds of node satisfy

$$a_u - \xi G_u \ge -\delta^2(Y_u - b_u). \tag{4.6}$$

A *record family* is deterministic. In both kinds of family the parent chord length R satisfies $L/2 < R \le L$, and its endpoints belong to their prescribed cubes. A good family consists of all ordered M-edge polygons joining these endpoints, whose first $M-1$ edges have length R/M, whose final edge has length in $[R/(2M), 3R/(2M)]$, and which satisfy the recorded child-endpoint restrictions, the bin restriction, and the conclusions of Lemma 3.2. A bad family consists of all parent and adverse chord pairs with the prescribed endpoint cubes, adverse chord length in $[R/(2M), 3R/(2M)]$, and adverse endpoints in the parent ball (3.2). It does not require the adverse chord to maximize a field-dependent quantity. The followed-child data are retained for counting and for the next node, but need not restrict the current Gaussian family further. Closing these parameter sets only increases the relevant suprema; the displayed lower bounds on chord lengths prevent degeneracy. Empty families are omitted.

Lemma 4.1 (Number of extensions)**.** *Given a parent scale and its two endpoint cubes, the number of possible next good records in bin k is at most*

$$C(k+1)^D, \tag{4.7}$$

where D depends only on d. The number of possible next bad records is at most $C\delta^{-D}$, after increasing D if necessary.

Proof. Choose the centers of the parent cubes as reference endpoints. By (3.7), a good-node vertex of index i lies within $C\delta L\sqrt{k+1}$ of the point at proportion i/M along the reference chord. The next grid has side between $\delta L/32$ and $\delta L/8$. A ball of this radius meets at most $C(k+1)^{d/2}$ next-grid cubes. Count two followed-child endpoints, M child choices, and three scale choices to get (4.7).

At a bad node, every recorded endpoint is within C_dL of the reference initial parent endpoint. Each adverse endpoint has at most $C\delta^{-d}$ parent-grid choices, and each followed endpoint has at most $C\delta^{-d}$ next-grid choices. Count the four endpoints and the three next scales. This bound does not count, or assume a bound for, child occurrences. □

Lemma 4.2 (Local Gaussian bounds)**.** *Uniformly in the parent location and scale and in N, a good record in bin k satisfies*

$$\mathbb{E}\sup Y_u \le C(k+1)^{1/4}, \qquad \sup \operatorname{Var} Y_u \le C\sqrt{k+1}. \tag{4.8}$$

For a bad record,

$$\mathbb{E}\sup Y_u \le C, \qquad \sup \operatorname{Var} Y_u \le C/\delta. \tag{4.9}$$

The suprema range over the indicated deterministic record family.

Proof. For a good node, Lemmas 2.1 and 3.2 give

$$\operatorname{Var} Y_u \le C\delta^{-1}\sqrt{a_u} \le C\sqrt{k+1}.$$

For the mean estimate we give a uniform parameterization, rather than take a union bound over a fine mesh of the field. Let q_i be the points at proportions i/M along the reference chord between the parent cube centers. Write the polygon vertices as

$$z_i = q_i + \delta L \eta_i, \qquad 0 \le i \le M.$$

Formula (3.7) places the parameter vector η in a ball of radius $C\sqrt{k+1}$ in a space of fixed dimension $p = d(M+1)$. Parent endpoints are among these parameters. Every edge length is bounded above and below by fixed positive multiples of L, since M is fixed.

For two members $\eta, \widetilde{\eta}$ of the family, Lemma 2.2 couples their normalized polygon measures within $C\delta L\|\eta - \widetilde{\eta}\|$. It couples the parent chord measures within the same bound. Both polygon measures have the growth bound at scale L. Apply Lemma 2.1 separately to these two differences and use $\operatorname{Var}(X+Z) \le 2\operatorname{Var} X + 2\operatorname{Var} Z$. The canonical distance of the rescaled variables satisfies

$$d_Y(\eta, \widetilde{\eta})^2 \le C\|\eta - \widetilde{\eta}\|. \tag{4.10}$$

Here and in this proof any fixed norm on the parameter space may be used. A Euclidean parameter net consequently gives

$$\mathcal{N}(T, d_Y, s) \le \left(1 + \frac{C\sqrt{k+1}}{s^2}\right)^p, \qquad \operatorname{diam}(T, d_Y) \le C(k+1)^{1/4}. \tag{4.11}$$

Nets for a subset can be chosen with centers in that subset by a maximal separated-set argument. Substitute $s = (k+1)^{1/4} r$ in (2.5). The resulting integral is bounded by $C(k+1)^{1/4}$, because $\int_0^C \sqrt{\log(2 + C/r^2)}\, dr < \infty$. A fixed reference variable in the family is centered. This proves the mean estimate without a logarithmic loss in k or δ.

For a bad record, both chord measures satisfy the growth bound at scale L, with a constant depending on M, and they can be coupled within CL. Thus $\operatorname{Var} Y_u \le C/\delta$. Within one record, the four endpoints vary in their prescribed cubes of side δL. After rescaling their displacements by δL, the parameter space is a fixed bounded set in $\mathbb{R}^{4d}$. The same stability and transport argument gives (4.10) and a bounded entropy integral. A deterministic reference chord pair has a centered Gaussian value, so its possibly large variance does not enter the expected supremum. This gives the first bound in (4.9). □

5 Covariance along a chain and the upper bound

For a fixed chain record, allow the geometry at different nodes to vary independently over its respective record families. This is an enlargement of the actual geometric choices, not a probabilistic independence assumption. Each signed chord or polygon measure is still paired with the same field H_N.

Lemma 5.1 (Chain variance)**.** *For every fixed chain record and every member of its enlarged product family,*

$$\operatorname{Var}\left(\sum_i Y_i\right) \le C \sum_{i \text{ good}} \sqrt{k_i+1} + \frac{C}{\delta}\#\{i : i \text{ bad}\}. \tag{5.1}$$

Proof. At a good node the two probability measures satisfy the growth bound at its chord scale R_i and are coupled within $CR_i\delta\sqrt{k_i+1}$. At a bad node the coupling displacement is at most CR_i, with the same type of growth bound. Define

$$v_i = \begin{cases} (k_i+1)^{1/4}, & i \text{ good}, \\ \delta^{-1/2}, & i \text{ bad}. \end{cases}$$

Let $L_i = 2^{-j_i}$ be the recorded dyadic scale. Applying Lemma 2.1 to a pair of nodes gives

$$|\operatorname{Cov}(Y_i, Y_l)| \le C 2^{-|j_i - j_l|/2} v_i v_l. \tag{5.2}$$

This estimate is uniform in the locations and in compatibility of the geometries. By (4.2), the j_i are distinct and separated by at least three. The matrix $2^{-|j_i-j_l|/2}$ has uniformly bounded row sums. Sum (5.2) and apply $2v_iv_l \le v_i^2 + v_l^2$ to obtain (5.1). □

Proposition 5.2 (Simultaneous path bound)**.** *There exist $C < \infty$ and $\xi_0 > 0$ such that for each fixed $\xi \in (0, \xi_0)$,*

$$\log D_N^{\xi} \ge -C\xi^{4/3}N - o_{\mathbb{P}}(N) \qquad \text{as } N \to \infty. \tag{5.3}$$

Here $o_{\mathbb{P}}(N)$ denotes a random error whose ratio to N tends to zero in probability, with ξ fixed.

Proof. For a fixed record of m internal nodes, let T be the supremum of $\sum_i Y_i$ over its enlarged product family. Pointwise the supremum of a sum over a product index set is the sum of the separate suprema, irrespective of correlations. Lemma 4.2 therefore bounds $\mathbb{E}T$ by the sum of the corresponding local means. Apply (2.6) with $t = 1$ and the variance bound of Lemma 5.1.

After subtracting the recorded costs b_i and summing over next records using Lemma 4.1, the contribution per good node is at most

$$Z_{\mathrm{g}} = C\sum_{k \ge 0} (k+1)^D \exp\{C(k+1)^{1/4} + C\sqrt{k+1} - k\} < \infty. \tag{5.4}$$

It is harmless to sum over bins which cannot occur when $a_u \le a_0$. The contribution per bad node is at most

$$Z_{\mathrm{b}}(\delta) = C\delta^{-D}\exp\{C + C/\delta - a_0/\delta^2\}. \tag{5.5}$$

Since $a_0 > 0$ is fixed, $Z_{\mathrm{b}}(\delta) \to 0$ as $\delta \downarrow 0$. Choose δ_0 so that $Z_{\mathrm{b}}(\delta) \le 1$ for $0 < \delta < \delta_0$.

The root scale and endpoint cubes are fixed. Given a deterministic partial record, the same numerical bounds count all extensions; no conditioning of Gaussian laws is performed. Let $\mathcal{R}_m$ be the countable set of possible deterministic chain records of length m with these root data. For

$\mathfrak{r} \in \mathcal{R}_m$, denote its Gaussian supremum by $T_{\mathfrak{r}}$ and its sum of recorded costs by $B_{\mathfrak{r}}$. Iterating the counting bounds gives, for every $T_0 > 0$,

$$\mathbb{P}\left[\exists\, m \le J_N,\ \mathfrak{r} \in \mathcal{R}_m : T_{\mathfrak{r}} - B_{\mathfrak{r}} > T_0\right] \le e^{-T_0} \sum_{m=1}^{J_N} (Z_{\mathrm{g}} + Z_{\mathrm{b}})^m. \tag{5.6}$$

The event on the left is a countable union of measurable Gaussian-supremum events. Its complement bounds every chain of every path, because the record families include all field-dependent geometries and, at bad nodes, all choices of the maximizing adverse child. No measurability assertion about an uncountable union of path events is needed.

Set

$$C_* = \log(2 + Z_{\mathrm{g}}) + 2, \qquad T_0 = C_* J_N.$$

Since $\sum_{m=1}^{J} z^m \le (1+z)^J$ for $z \ge 0$, the right side of (5.6) is at most e^{-2J_N}. On the complementary event, every chain of every path has $\sum_i (Y_i - b_i) \le C_* J_N$. Average this inequality over the leaf distribution of the unit flow. The contribution of a node to this average is its flow $\theta(u)$. Hence (4.6) and Lemma 3.1 give, simultaneously for every path,

$$\log \int_P e^{\xi H_N} |dz| \ge \xi H_0 - C_* \delta^2 J_N - \xi \omega_N.$$

Take the infimum over paths. By (3.3), $J_N \le CN + C$, and by Lemma 2.3 the other terms are $o_{\mathbb{P}}(N)$ for fixed ξ. Since $\delta^2 = \xi^{4/3}$, this proves (5.3). □

Proof of Theorem 1.1. Fix $\xi \in (0, \xi_0)$ and $\eta > 0$. Proposition 5.2 gives, with probability tending to one,

$$D_N^{\xi} \ge \exp\{-(C\xi^{4/3} + \eta)N\}.$$

For all sufficiently large N, this lower threshold lies below every median. Thus (1.3) implies $\lambda_d(\xi) \le (C\xi^{4/3} + \eta)/\log 2$. Let $\eta \downarrow 0$ and absorb $1/\log 2$ into C_d.

The planar estimate of [3], in the normalization used here, is $\lambda_2(\xi) \ge c\xi^{4/3}/\log(1/\xi)$ for small ξ. This form is also stated immediately after [1, Theorem 1.2]. Appendix A proves $\lambda_d(\xi) \ge \lambda_2(\xi)$ for the point-to-point exponents (1.3). This proves (1.4). If $L = \log(1/\xi)$, these bounds give

$$\log \lambda_d(\xi) = -\tfrac{4}{3}L + O(\log L),$$

which proves (1.5). The order of limits throughout is $N \to \infty$ for fixed ξ, followed by $\xi \downarrow 0$. □

Proof of Corollary 1.2. As $\gamma \downarrow 0$, (1.6) and the known bound $Q_d(\xi) \le \xi^{-1} + \sqrt{2}$ from [4, Proposition 1.1] imply $\xi \to 0$. Equations (1.3) and (1.6) give the exact identity

$$\mathsf{d}_\gamma - d = \frac{\gamma^2}{2} + \mathsf{d}_\gamma \lambda_d(\gamma/\mathsf{d}_\gamma). \tag{5.7}$$

Equivalently, $\mathsf{d}_\gamma = (d + \gamma^2/2)/(1 - \lambda_d(\xi))$. Theorem 1.1 therefore implies $\mathsf{d}_\gamma \to d$ and $\xi \sim \gamma/d$. Substitute the two bounds from (1.4) into (5.7). Since $\log(1/\xi) \sim \log(1/\gamma)$ and $\gamma^2 = o(\gamma^{4/3}/\log(1/\gamma))$, the bounds (1.7) follow after adjusting constants. Taking logarithms gives (1.8). □

6 The transverse Gaussian limit and a separate coefficient question

This section is not needed for Theorem 1.1. It identifies the dimension-dependent Gaussian problem suggested by a small transverse perturbation of a line. Let h now denote the canonical whole-space log-correlated field, with covariance on zero-mass measures given by the kernel $\log|x-y|^{-1}$. Put $m = d-1$, and for a continuous function $f : [0,1] \to \mathbb{R}^m$ define

$$\mu_{\delta f} = \int_0^1 \delta_{(t,\delta f(t))}\, dt.$$

These graph measures satisfy a one-dimensional growth bound, since their projection to the first coordinate is Lebesgue measure. They have finite logarithmic energy on compact sets, so zero-mass differences can be paired with h.

Proposition 6.1 (Transverse covariance)**.** *For continuous $f, g : [0,1] \to \mathbb{R}^m$,*

$$\lim_{\delta \downarrow 0} \delta^{-1} \operatorname{Var}\langle h, \mu_{\delta f} - \mu_{\delta g}\rangle = 2\pi \int_0^1 |f(t) - g(t)|\, dt. \tag{6.1}$$

Consequently, for any finite family of such functions, the centered Gaussian variables $\delta^{-1/2}\langle h, \mu_{\delta f} - \mu_0\rangle$ converge jointly to a centered Gaussian process $\mathcal{G}_d(f)$ with $\mathcal{G}_d(0) = 0$ and covariance

$$\mathbb{E}[\mathcal{G}_d(f)\mathcal{G}_d(g)] = \pi \int_0^1 \left(|f(t)| + |g(t)| - |f(t) - g(t)|\right) dt. \tag{6.2}$$

Proof. Expand the variance as a double integral of four logarithmic kernels. Set $s = t + \delta u$. The common $\log\delta$ and $\log|u|$ terms cancel. The integrand of the resulting expression for the variance divided by δ is

$$\begin{aligned} &\tfrac{1}{2}\log\left(1 + \frac{|f(t) - g(t+\delta u)|^2}{u^2}\right) + \tfrac{1}{2}\log\left(1 + \frac{|g(t) - f(t+\delta u)|^2}{u^2}\right) \\ &\quad - \tfrac{1}{2}\log\left(1 + \frac{|f(t) - f(t+\delta u)|^2}{u^2}\right) - \tfrac{1}{2}\log\left(1 + \frac{|g(t) - g(t+\delta u)|^2}{u^2}\right), \end{aligned}$$

with $0 < t < 1$ and $0 < t + \delta u < 1$. For $u \neq 0$ it converges to $\log(1 + |f(t) - g(t)|^2/u^2)$. If B bounds $|f|$ and $|g|$, its absolute value is bounded by $2\log(1 + 4B^2/u^2)$, an integrable function of u. Dominated convergence and the identity

$$\int_{\mathbb{R}} \log(1 + a^2/u^2)\, du = 2\pi|a|$$

prove (6.1). For completeness, differentiate the integral in $a > 0$ to obtain 2π, and use its limit zero as $a \downarrow 0$. Polarization proves (6.2) and joint convergence. □

The covariance in (6.2) is positive semidefinite by the proposition. On each finite-dimensional space of piecewise affine functions, its canonical distance is bounded by a constant times the square root of the uniform distance. It consequently has a separable continuous version on compact parameter sets. This supplies all processes used below without any appeal to a pointwise random potential.

For a fixed piecewise affine f with $f(0) = f(1) = 0$,

$$\operatorname{len}\{(t, \delta f(t)) : 0 \le t \le 1\} = 1 + \delta^2 \mathcal{E}_d(f) + O_f(\delta^4), \qquad \mathcal{E}_d(f) = \frac{1}{2}\int_0^1 |f'(t)|^2\, dt. \tag{6.3}$$

Thus the length penalty has size δ^2 and the linear Gaussian term has size $\xi\sqrt{\delta}$. Their balance gives $\delta = \xi^{2/3}$. This calculation explains the power but is not used as a uniform Taylor expansion of LFPP: H_N has variance $N\log 2$, and no such expansion is uniform in N.

6.1 A subadditive Gaussian quantity

Let $V_n^{(m)}$ be the space of continuous $\mathbb{R}^m$-valued functions on $[0,1]$ which vanish at the endpoints and are affine on the dyadic intervals of length 2^{-n}. Set

$$A_{n,d} = \mathbb{E} \sup_{f \in V_n^{(d-1)}} \{\mathcal{G}_d(f) - \mathcal{E}_d(f)\}. \tag{6.4}$$

Proposition 6.2. *For $n, r \geq 1$, the quantities $A_{n,d}$ are finite and satisfy*

$$A_{n+r,d} \leq A_{n,d} + A_{r,d}. \tag{6.5}$$

For $a, b > 0$,

$$\mathbb{E} \sup_{f \in V_n^{(d-1)}} \{a\mathcal{G}_d(f) - b\mathcal{E}_d(f)\} = a^{4/3} b^{-1/3} A_{n,d}. \tag{6.6}$$

In particular,

$$\kappa_d^{\mathrm{var}} = \lim_{n\to\infty} \frac{A_{n,d}}{n \log 2} = \inf_{n \geq 1} \frac{A_{n,d}}{n \log 2} \in [0, \infty) \tag{6.7}$$

exists.

Proof. On the energy ball $\{\mathcal{E}_d(f) \leq R\}$, the variance is at most $C\sqrt{R}$ by the one-dimensional Poincaré inequality. The finite-dimensional entropy integral is finite. Vertical scaling gives $(\mathcal{G}_d(cf))_f \overset{d}{=} (\sqrt{c}\,\mathcal{G}_d(f))_f$ for $c > 0$, so the expected supremum on this ball is $R^{1/4}$ times that on the unit ball. Gaussian concentration on the shells $j \leq \mathcal{E}_d(f) < j+1$ implies integrability of the supremum after subtracting the energy: for large j, overcoming a fixed positive multiple of j has probability at most $C_{n,d} \exp(-c_{n,d} j^{3/2})$. This proves finiteness. The substitution $f = (a/b)^{2/3} g$ proves (6.6).

For $f \in V_{n+r}^{(m)}$, let g be its interpolant on the coarse dyadic mesh and set $v = f - g$. Orthogonality on each coarse interval gives $\mathcal{E}_d(f) = \mathcal{E}_d(g) + \mathcal{E}_d(v)$. For two decompositions,

$$\mathbb{E}[(\mathcal{G}_d(g+v) - \mathcal{G}_d(g'+v'))^2] \leq 2\pi \int |g - g'| + 2\pi \int |v - v'|.$$

The right side is the increment variance of $\mathcal{G}_d^{(1)}(g) + \mathcal{G}_d^{(2)}(v)$ for independent copies. Apply the Sudakov–Fernique inequality with the same deterministic energy penalty on both sides; this form follows by the usual Gaussian interpolation, also with nonconstant deterministic means, as in [7]. The comparison supremum separates.

On a coarse interval of length $\ell = 2^{-n}$, write $v(a + \ell t) = \ell u(t)$. Both the Gaussian process and the energy are multiplied in law by ℓ. Contributions on disjoint coarse intervals are independent for this comparison process, by (6.2). The fine expected supremum is therefore $2^n \ell A_{r,d} = A_{r,d}$, while the coarse one is $A_{n,d}$. This proves (6.5). Finite approximations to energy-bounded sets followed by the integrability just proved justify all comparisons. Finally apply Fekete's lemma. □

Proposition 6.3 (Comparison between dimensions)**.** *For $d \geq 2$ and every $n \geq 1$,*

$$(d-1)^{2/3} A_{n,2} \leq A_{n,d} \leq (d-1) A_{n,2}. \tag{6.8}$$

Consequently,

$$(d-1)^{2/3} \kappa_2^{\mathrm{var}} \leq \kappa_d^{\mathrm{var}} \leq (d-1) \kappa_2^{\mathrm{var}}. \tag{6.9}$$

Proof. Put $m = d-1$ and let $Y(f) = \sum_{j=1}^{m} \mathcal{G}_2^{(j)}(f_j)$ for independent scalar copies. The inequalities $m^{-1/2}\sum_j |z_j| \le |z| \le \sum_j |z_j|$ show that the increment variances of $\mathcal{G}_d$ lie between those of $m^{-1/4}Y$ and Y. Apply Sudakov–Fernique with the common penalty $\mathcal{E}_d(f) = \sum_j \mathcal{E}_2(f_j)$. The coordinate suprema separate. By (6.6), their expectations are $m \cdot m^{-1/3} A_{n,2}$ and $mA_{n,2}$, respectively. This proves (6.8); divide by $n \log 2$ and take limits. □

Conjecture 6.4 (Leading coefficient; not a conclusion of this manuscript)**.** For every fixed $d \ge 2$,

$$\lambda_d(\xi) = \kappa_d^{\mathrm{var}} \xi^{4/3} + o(\xi^{4/3}) \quad \text{as } \xi \downarrow 0, \qquad \kappa_d^{\mathrm{var}} > 0. \tag{6.10}$$

The finite-dimensional Gaussian convergence does not prove this conjecture. It does not justify an interchange between the small-parameter limit and arbitrarily many LFPP scales. In particular, the fixed subdivision parameter in Sections 3–5 gives only an upper constant, not its optimal value. A leading-coefficient identification would require a sharp large-block upper comparison and a constructive comparison controlling expected weighted occupation measures uniformly over all subsequent refinements.

The scalar Brownian-potential variational problem is related to the work of Otto–Palmieri [8]. Their result concerns the variational action, not the LFPP comparison in (6.10). Neither a scalar action homogenization theorem nor Proposition 6.2 by itself identifies the LFPP coefficient. If (6.10) holds, (5.7) would imply

$$\mathsf{d}_\gamma = d + d^{-1/3} \kappa_d^{\mathrm{var}} \gamma^{4/3} + o(\gamma^{4/3}).$$

This last display is conditional and is not part of Corollary 1.2.

A Cutoff compatibility and restriction to a plane

We supply the regularization comparison used in the lower bound, so that a set-to-set exponent is not silently substituted for the point-to-point exponent (1.3).

Lemma A.1 (A cutoff compared with a subspace mollification)**.** *Let $1 \le k \le d$ and identify $\mathbb{R}^k$ with $\mathbb{R}^k \times \{0\}^{d-k}$. Let $\rho \in C_c^\infty(\mathbb{R}^k)$ be a nonnegative, radially symmetric probability density, and set $\rho_\varepsilon(x) = \varepsilon^{-k}\rho(x/\varepsilon)$. There is a coupling of $H_N|_{\mathbb{R}^k}$ with a canonical k-dimensional log-correlated Gaussian field $h^{(k)}$, with a fixed normalization, such that for each compact $V \subset \mathbb{R}^k$,*

$$\sup_{x \in V} |H_N(x) - (h^{(k)} * \rho_{2^{-N}})(x)| = O_{\mathbb{P}}(\sqrt{N}). \tag{A.1}$$

Proof. Use the same noise as in (1.1), now at all scales. Pair it with $K_t * \sigma$, where σ is a smooth compactly supported zero-mass measure on the subspace. This defines a centered Gaussian distribution on the subspace. To identify its covariance, write $C(z) = (\kappa * \kappa)(z)$, which is radial, smooth and compactly supported, with $C(0) = 1$. For $r, s > 0$,

$$\int_0^\infty \big(C(re_1/t) - C(se_1/t)\big)\frac{dt}{t} = \log(s/r). \tag{A.2}$$

For example, truncate the integral at both ends and change variables; the difference is determined by the endpoint values $C(0) = 1$ and $C(\infty) = 0$. It follows that the covariance on zero-mass measures is precisely the logarithmic covariance. Choose its additive constant by setting its pairing with the subspace measure $\rho(x)\,dx$ equal to zero. This constructs $h^{(k)}$.

Let $\mu_{\varepsilon,x}$ denote $\rho_\varepsilon(\cdot - x)\,dx$ embedded in the subspace, and let $\mu_0 = \rho\,dx$. Then

$$(h^{(k)} * \rho_\varepsilon)(x) = \int_0^\infty \int_{\mathbb{R}^d} K_t * (\mu_{\varepsilon,x} - \mu_0)(y)\, W(\,dy,\,dt),$$

where symmetry of ρ and κ fixes the convolution convention.

Write $\varepsilon = 2^{-N}$. At scales $t < \varepsilon$, the k-dimensional mass bound for $\mu_{\varepsilon,x}$ and compact support of C give

$$\|K_t * \mu_{\varepsilon,x}\|_2^2 \le Ct^{-1}(t/\varepsilon)^k.$$

Its integral over $(0, \varepsilon)$ is bounded independently of ε. At scales $\varepsilon \le t \le 1$, the displacement of the mollifier and the bound $\|\nabla K_t\|_2 \le Ct^{-3/2}$ give

$$\|K_t * \mu_{\varepsilon,x} - K_t(x - \cdot)\|_2 \le C\varepsilon t^{-3/2},$$

whose square is integrable with bounded integral. The normalization measure contributes a bounded squared norm on $(0, 1)$. Finally, for $t \ge 1$, zero total mass and bounded first moments, uniformly for $x \in V$, give

$$\|K_t * (\mu_{\varepsilon,x} - \mu_0)\|_2 \le C_V t^{-3/2}.$$

The point variance of the error in (A.1) is therefore uniformly bounded.

The derivative variance of H_N in every subspace direction is at most $C\varepsilon^{-2}$. The derivative variance of $h^{(k)} * \rho_\varepsilon$ has the same bound: a derivative of the mollifier has total mass zero, and scaling the logarithmic covariance shows that its variance is ε^{-2} times a fixed finite constant. Thus the error process has canonical increments at most $C \min(1, |x - y|/\varepsilon)$. Its canonical covering numbers on V are at most $C_V \varepsilon^{-k} s^{-k}$ for $0 < s < 1$. The entropy integral and Markov's inequality prove (A.1). □

Lemma A.2 (Concentration at the exponent scale)**.** *For a fixed $\xi > 0$, the cutoff distance in* (1.2) *satisfies*

$$\log D_N^\xi - \log a_N(\xi) = O_{\mathbb{P}}(\sqrt{N}). \tag{A.3}$$

Consequently $-\log D_N^\xi/(N \log 2) \to \lambda_d(\xi)$ in probability.

Proof. For a Cameron–Martin shift f of the driving white noise, the corresponding change of $H_N(x)$ is at most $\sqrt{N \log 2}\|f\|_2$, uniformly in x, by Cauchy–Schwarz. Comparing every path cost under the two fields shows that $\log D_N^\xi$ is $\xi\sqrt{N \log 2}$-Lipschitz on Gaussian space. Gaussian concentration gives fluctuations of order $\xi\sqrt{N}$ about its mean, and hence also about any median. The mean is finite since $\xi \inf_{\overline{U}} H_N \le \log D_N^\xi \le \xi \sup_{[0,e_1]} H_N$, and these smooth Gaussian suprema have finite expectations. The logarithm of any median of D_N^ξ is a median of its logarithm. This proves (A.3); now use (1.3). □

Proposition A.3 (Restriction comparison)**.** *For integers $d \ge k \ge 2$ and every $\xi > 0$,*

$$\lambda_d(\xi) \ge \lambda_k(\xi). \tag{A.4}$$

Proof. Restrict the paths in (1.2) to the coordinate subspace $\mathbb{R}^k$. This can only increase the distance. By Lemma A.1, the logarithm of the restricted distance differs by $O_{\mathbb{P}}(\sqrt{N})$ from that defined by a smooth mollification of the canonical k-dimensional field. Apply the same lemma in dimension k to compare the latter with its own cutoff approximation. The fields can be coupled through their common mollified-field law; alternatively, the two comparisons give the same convergence in distribution to a deterministic exponent. Lemma A.2 identifies that exponent as $\lambda_k(\xi)$. The full distance has exponent $\lambda_d(\xi)$ by the same lemma. Divide the pathwise distance inequality by the negative scale $-N \log 2$ to obtain (A.4). □

Taking $k = d$ and comparing two kernels through the same canonical mollification also proves kernel independence of the exponent.

For completeness, the circle-average regularization in the planar input can be compared in the same coupling. Let $\sigma_{\varepsilon,x}$ be uniform measure on the circle of radius ε about $x \in \mathbb{R}^2$, and let $\mu_{\varepsilon,x}$ be the planar smooth mollifier measure above. Both satisfy (2.1) at scale ε and can be coupled within $C\varepsilon$. The proof of Lemma 2.1 integrates over all $t > 0$, so it applies unchanged to canonical zero-mass field pairings. Hence

$$E_\varepsilon(x) := \langle h^{(2)}, \sigma_{\varepsilon,x} - \mu_{\varepsilon,x} \rangle \quad \text{satisfies} \quad \operatorname{Var} E_\varepsilon(x) \le C, \qquad \mathbb{E}[(E_\varepsilon(x) - E_\varepsilon(y))^2] \le C \min(1, |x-y|/\varepsilon).$$

For the increment bound, apply the transport estimate separately to each measure and its translate. A parameter mesh of size $c\varepsilon s^2$ gives a canonical mesh of size s, so the entropy integral gives $\sup_{x \in V} |E_\varepsilon(x)| = O_{\mathbb{P}}(\sqrt{\log(1/\varepsilon)})$ on each compact V. Thus circle averaging and smooth mollification yield the same distance exponent.

For $k = 2$, the canonical field is the whole-plane GFF with logarithmic coefficient one. On a compact subset of a larger domain it can be coupled to a zero-boundary GFF by adding a continuous Gaussian function, so this change contributes only $O_{\mathbb{P}}(1)$ to the logarithmic distance. Apply [3, Theorem 1.1] to paths in a fixed square inside U containing 0 and e_1; a fixed affine rescaling changes no exponent. Their expected-distance upper bound implies the corresponding median-exponent lower bound by Markov's inequality. The cutoff and circle-average comparisons just proved transfer it to (1.3). This supplies the planar input, with the normalization and form also stated in [1].

B Three internal audit passes and the novelty check

The checks below were performed in sequence during preparation of this manuscript. They are internal checks of the argument, not independent refereeing or a formal proof certificate.

Pass 1: local estimates and deterministic geometry

The proof was reconstructed from the cutoff covariance, rather than assuming that every planar GFF identity survives in higher dimension. The transport estimate was proved directly from the compactly supported noise kernel. This avoids an unstated change to a Gaussian kernel outside the precise setup of [4]. The root containment constant was made dimension-dependent: the cube $(-2, 2)^d$ is not contained in a Euclidean ball of radius 4 in all dimensions. Completed pieces, retained residuals, merged residuals, and zero-chord terminal returns were checked separately. The good-node child count and angle bound follow from explicit inequalities in Lemma 3.2. The normalized-arclength comparison was supplied as Lemma 2.2; vertex displacement alone is not a justification without this step.

Pass 2: external inputs and uniformity over paths

The definition of the exponent was matched to the point-to-point median of [4, Proposition 1.1], not to an unrestricted set-to-set exponent in another phase. Appendix A now proves the required subspace and cutoff comparisons, including the circle-average regularization in the planar input. The only non-Gaussian external input for the two-sided bound, besides existence of the exponent, is the published planar lower bound. The union bound is over deterministic record families, including incompatible geometries, so it covers field-dependent paths without conditioning on a chosen geodesic. Bad nodes do not require a bound on the number of their children: only the adverse chord and the followed chord are recorded. The deterministic families retain explicit lower bounds on every chord

length, including the adverse chord, even when that length is not a separate item in the record. Correlations are retained and estimated by (5.2). No independence of different nodes is used.

Pass 3: independent reconstruction of the final implication

The entropy calculation was repeated after scaling the parameter ball: the substitution in (4.11) gives a fourth-root mean bound, with no hidden $\log(1/\delta)$ term. Summing the variances gives a square-root bin cost, which is absorbed by the linear penalty in (5.4). At bad nodes, the penalty is of order δ^{-2}, whereas the concentration term is only of order δ^{-1}; hence (5.5) tends to zero. The union event was rewritten explicitly as a countable union over deterministic records, rather than assigning a probability to an uncountable union over paths. The chain bound was then averaged with the unit flow and reinserted into the exact Jensen inequality. The sign in passing to the distance exponent and the order $N \to \infty$ before $\xi \downarrow 0$ were checked explicitly. Finally, the coefficient assertion was separated from the theorem: the transverse limit and subadditivity do not identify the LFPP coefficient.

Literature and priority

The literature check was made on September 23, 2026. The sources checked include [4, 1, 2, 3] and the related Gaussian-action work [8]. In the retrieved text, [4, Problem 7.4] asks for the small-parameter asymptotics, and [1, Theorem 1.3] supplies only the linear upper bound discussed in Section 1. The estimate $\lambda_d(\xi) \le C_d \xi^{4/3}$ was not found in these sources or in the targeted searches. The work [2] supplies the higher-dimensional metric setup and dimension correspondence, not this small-parameter estimate. This supports treating the upper-bound argument as a potentially new result requiring independent scrutiny, not asserting established priority. None of the three internal passes is a substitute for such scrutiny.

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